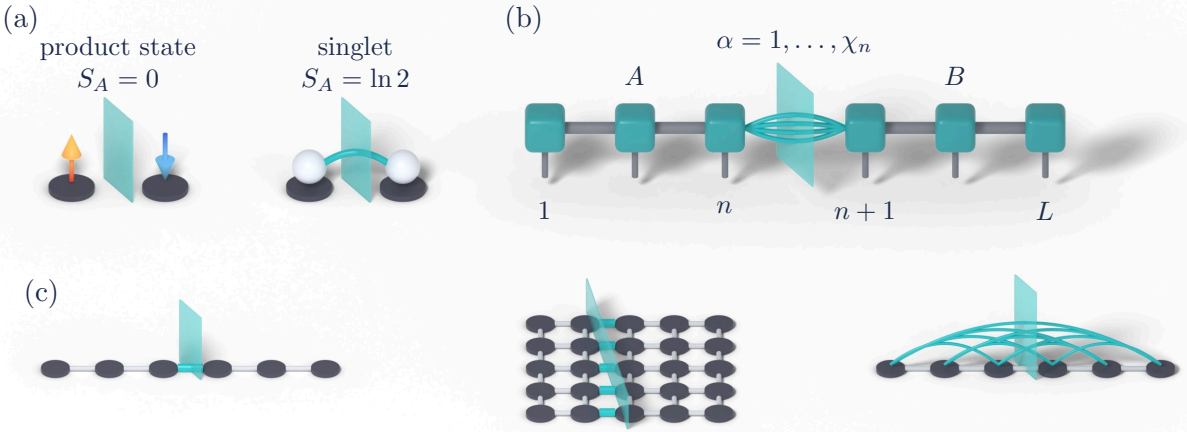

(a)
product state
$S_A = 0$
singlet
$S_A = \ln 2$
(b)
$\alpha = 1, \ldots, \chi_n$
A
B
1
n
$n+1$
L
(c)
chain: one bond
square lattice: a line of bonds
random state: volume law